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# Association Rule Learning on the Groceries Dataset

*Discovering Product Co-Purchase Patterns with the Apriori Algorithm*

**Dataset:** Groceries | **Software:** R (arules, arulesViz)

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**Course:** Data mining(Unsupervised learning methods)  
**Assignment:** Association Rule Learning  
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## Executive Summary

This report applies association rule learning to the classic **Groceries** transaction dataset (9,835 shopping baskets, 169 distinct items) to uncover which products customers tend to purchase together. Using the Apriori algorithm implemented in R's **arules** package, frequent itemsets and association rules were mined at a minimum support of 1% and minimum confidence of 30%, yielding 125 qualifying rules from an initial itemset search.

The results show that *whole milk* and *other vegetables* are the most frequently purchased individual items and also anchor the highest-support product pairing ( $\{\text{other vegetables, whole milk}\}$ , support  $\approx 7.5\%$ ). However, the strongest *associations* as measured by lift involve less common but more tightly linked items such as *citrus fruit*, *root vegetables*, and *beef*, several of which showed lift values above 3.0, indicating purchases roughly three times more likely than chance would predict. The rule  $\{\text{citrus fruit, other vegetables}\} \Rightarrow \{\text{root vegetables}\}$  achieved the highest lift (3.30), while  $\{\text{citrus fruit, root vegetables}\} \Rightarrow \{\text{other vegetables}\}$  achieved the highest confidence (58.6%). This report walks through the full analysis, answers all eight assignment questions in detail, and closes with concrete merchandising recommendations grounded in the discovered rules.

## Table of Contents

|                              |    |
|------------------------------|----|
| Objective                    | 3  |
| 1 Understand the Data        | 4  |
| 2 Frequent Itemsets          | 5  |
| 3 Association Rules          | 6  |
| 4 Understanding the Measures | 9  |
| 5 Find Interesting Rules     | 10 |
| 6 Interpret a Rule           | 11 |
| 7 Business Application       | 11 |
| 8 Understanding Association  | 12 |
| 9 Conclusion                 | 12 |
| 10 Appendix: Full R Code     | 12 |

## Objective

The purpose of this assignment is to understand how association rule learning can be used to discover relationships between products that customers purchase together. Using the **Groceries** dataset, an association rule analysis was performed in R, covering data exploration, frequent itemset mining, rule generation, interpretation of support/confidence/lift, and a business application grounded in the discovered patterns.

## Methodology

The analysis was performed in **R** using the **arules** package. The Groceries data was loaded in basket (transaction) format, where each row represents one shopping trip and contains a variable-length list of purchased items. The **apriori()** function was used twice: first to mine *frequent itemsets* (minimum support = 0.01, minimum length = 2 items), and second to generate *association rules* (minimum support = 0.01, minimum confidence = 0.30, minimum length = 2 items). Results were sorted and inspected using **sort()**, **inspect()**, and converted to data frames with **kable()** for tabular reporting. Visualizations were produced with **itemFrequencyPlot()** and the **arulesViz** package (scatter and network graph views).

```
library(arules)
library(arulesViz)
library(knitr)
library(RColorBrewer)

# Load from groceries.csv (basket-format, no header,
# variable number of items per row).
transactions <- read.transactions("groceries.csv", format = "basket",
                                   sep = ",", rm.duplicates = TRUE)

summary(transactions)
```

## Dataset summary (R output):

```
transactions as itemMatrix in sparse format with
9835 rows (elements/itemsets/transactions) and
169 columns (items) and a density of 0.02609146
```

most frequent items:

|            |                  |            |      |
|------------|------------------|------------|------|
| whole milk | other vegetables | rolls/buns | soda |
| 2513       | 1903             | 1809       | 1715 |
| yogurt     | (Other)          |            |      |
| 1372       | 34055            |            |      |

element (itemset/transaction) length distribution:

| sizes |      |      |      |     |     |     |     |     |     |     |     |    |    |    |    |
|-------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-----|----|----|----|----|
| 1     | 2    | 3    | 4    | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13 | 14 | 15 | 16 |
| 2159  | 1643 | 1299 | 1005 | 855 | 645 | 545 | 438 | 350 | 246 | 182 | 117 | 78 | 77 | 55 | 46 |
| 17    | 18   | 19   | 20   | 21  | 22  | 23  | 24  | 26  | 27  | 28  | 29  | 32 |    |    |    |
| 29    | 14   | 14   | 9    | 11  | 4   | 6   | 1   | 1   | 1   | 1   | 3   | 1  |    |    |    |

|       |         |        |       |         |        |
|-------|---------|--------|-------|---------|--------|
| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.   |
| 1.000 | 2.000   | 3.000  | 4.409 | 6.000   | 32.000 |

## 1 Understand the Data

```
n_transactions <- length(transactions)
n_items <- ncol(transactions)

n_transactions # [1] 9835
n_items       # [1] 169

top10 <- sort(itemFrequency(transactions, type = "relative"), decreasing = TRUE)[1:10]
kable(data.frame(Item = names(top10), Support = round(as.numeric(top10), 4)))

itemFrequencyPlot(transactions, topN = 10, type = "relative",
                  main = "Top 10 Most Frequent Items",
                  col = brewer.pal(8, "Pastel2"))
```

| Item             | Support |
|------------------|---------|
| whole milk       | 0.2555  |
| other vegetables | 0.1935  |
| rolls/buns       | 0.1839  |
| soda             | 0.1744  |
| yogurt           | 0.1395  |
| bottled water    | 0.1105  |
| root vegetables  | 0.1090  |
| tropical fruit   | 0.1049  |
| shopping bags    | 0.0985  |
| sausage          | 0.0940  |

Table 1: Top 10 most frequently purchased items (relative support)

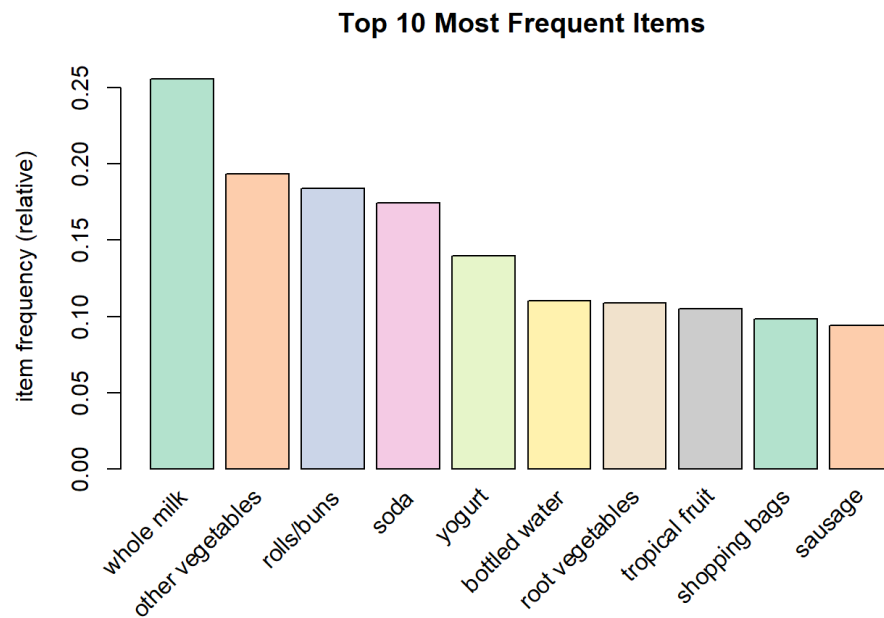


Figure 1: Relative item frequency for the top 10 most purchased grocery items

### Answer

- a. Number of transactions:** The dataset contains **9,835** transactions (shopping baskets).
- b. Number of distinct items:** The dataset contains **169** unique grocery items (columns of the sparse item matrix).
- c. 10 most frequently purchased items:** Shown in Table 1 and Figure 1. *Whole milk*, *other vegetables*, and *rolls/buns* lead by a wide margin, each appearing in roughly 18-26% of all transactions, consistent with these being everyday staple items that nearly every household buys regardless of the rest of their basket.

## 2 Frequent Itemsets

```
itemsets <- apriori(transactions, parameter = list(
  target = "frequent itemsets",
  support = 0.01,
  minlen = 2
), control = list(verbose = FALSE))

itemsets_sorted <- sort(itemsets, by = "support", decreasing = TRUE)
itemsets_df <- as(itemsets_sorted, "data.frame")
kable(head(itemsets_df, 10))
```

| ID  | Itemset                             | Support   | Count |
|-----|-------------------------------------|-----------|-------|
| 213 | {other vegetables, whole milk}      | 0.0748348 | 736   |
| 212 | {rolls/buns, whole milk}            | 0.0566345 | 557   |
| 210 | {whole milk, yogurt}                | 0.0560244 | 551   |
| 203 | {root vegetables, whole milk}       | 0.0489070 | 481   |
| 202 | {other vegetables, root vegetables} | 0.0473818 | 466   |
| 209 | {other vegetables, yogurt}          | 0.0434164 | 427   |
| 211 | {other vegetables, rolls/buns}      | 0.0426029 | 419   |
| 198 | {tropical fruit, whole milk}        | 0.0422979 | 416   |
| 207 | {soda, whole milk}                  | 0.0400610 | 394   |
| 205 | {rolls/buns, soda}                  | 0.0383325 | 377   |

Table 2: Top 10 frequent itemsets (2+ items, support threshold = 1%), sorted by support

### Answer

**a-b. Frequent itemsets and support values:** At least 10 frequent itemsets containing two or more products are listed in Table 2 (support threshold = 1%), together with their support values.

**c. Highest-support itemset:** {other vegetables, whole milk}, with support  $\approx 0.075$  (about 7.5% of all transactions contain both items).

**d. Interpretation of support:** The support value tells us how common a combination of products is across *all* transactions. A support of  $\approx 0.075$  for {other vegetables, whole milk} means nearly 1 in 13 shopping trips include both items together these are two of the most habitually co-purchased staples in the store. Itemsets with very low support represent rare combinations not worth building marketing decisions around, while high-support itemsets point to strong, everyday purchasing patterns worth acting on (e.g., joint promotions, adjacent shelf placement, or bundled pricing).

## 3 Association Rules

```
rules <- apriori(transactions, parameter = list(
  support = 0.01,
  confidence = 0.3,
  minlen = 2
), control = list(verbose = FALSE))

length(rules) # [1] 125

rules_by_lift <- sort(rules, by = "lift", decreasing = TRUE)
rules_df <- as(rules_by_lift, "data.frame")
kable(head(rules_df, 10))

# Export full rule table for submission
write.csv(rules_df, "association_rules_output.csv", row.names = FALSE)

plot(rules, measure = c("support", "confidence"), shading = "lift",
```

```

main = "Association Rules: Support vs Confidence (shaded by Lift)")

plot(head(rules_by_lift, 10), method = "graph",
     main = "Top 10 Rules by Lift - Network Graph")

```

A total of **125 association rules** met the minimum support (1%) and minimum confidence (30%) thresholds. The top 10, ranked by lift, are shown in Table 3.

| Rule (Antecedent $\Rightarrow$ Consequent)                         | Supp.  | Conf.        | Cov.   | Lift         | Count |
|--------------------------------------------------------------------|--------|--------------|--------|--------------|-------|
| {citrus fruit, other vegetables} $\Rightarrow$ {root vegetables}   | 0.0104 | 0.359        | 0.0289 | <b>3.295</b> | 102   |
| {other vegetables, tropical fruit} $\Rightarrow$ {root vegetables} | 0.0123 | 0.343        | 0.0359 | 3.145        | 121   |
| {beef} $\Rightarrow$ {root vegetables}                             | 0.0174 | 0.331        | 0.0525 | 3.040        | 171   |
| {citrus fruit, root vegetables} $\Rightarrow$ {other vegetables}   | 0.0104 | <b>0.586</b> | 0.0177 | 3.030        | 102   |
| {root vegetables, tropical fruit} $\Rightarrow$ {other vegetables} | 0.0123 | 0.585        | 0.0210 | 3.021        | 121   |
| {other vegetables, whole milk} $\Rightarrow$ {root vegetables}     | 0.0232 | 0.310        | 0.0748 | 2.842        | 228   |
| {curd, whole milk} $\Rightarrow$ {yogurt}                          | 0.0101 | 0.385        | 0.0261 | 2.761        | 99    |
| {rolls/buns, root vegetables} $\Rightarrow$ {other vegetables}     | 0.0122 | 0.502        | 0.0243 | 2.595        | 120   |
| {root vegetables, yogurt} $\Rightarrow$ {other vegetables}         | 0.0129 | 0.500        | 0.0258 | 2.584        | 127   |
| {tropical fruit, whole milk} $\Rightarrow$ {yogurt}                | 0.0152 | 0.358        | 0.0423 | 2.568        | 149   |

Table 3: Top 10 association rules ranked by lift (support  $\geq 0.01$ , confidence  $\geq 0.30$ )

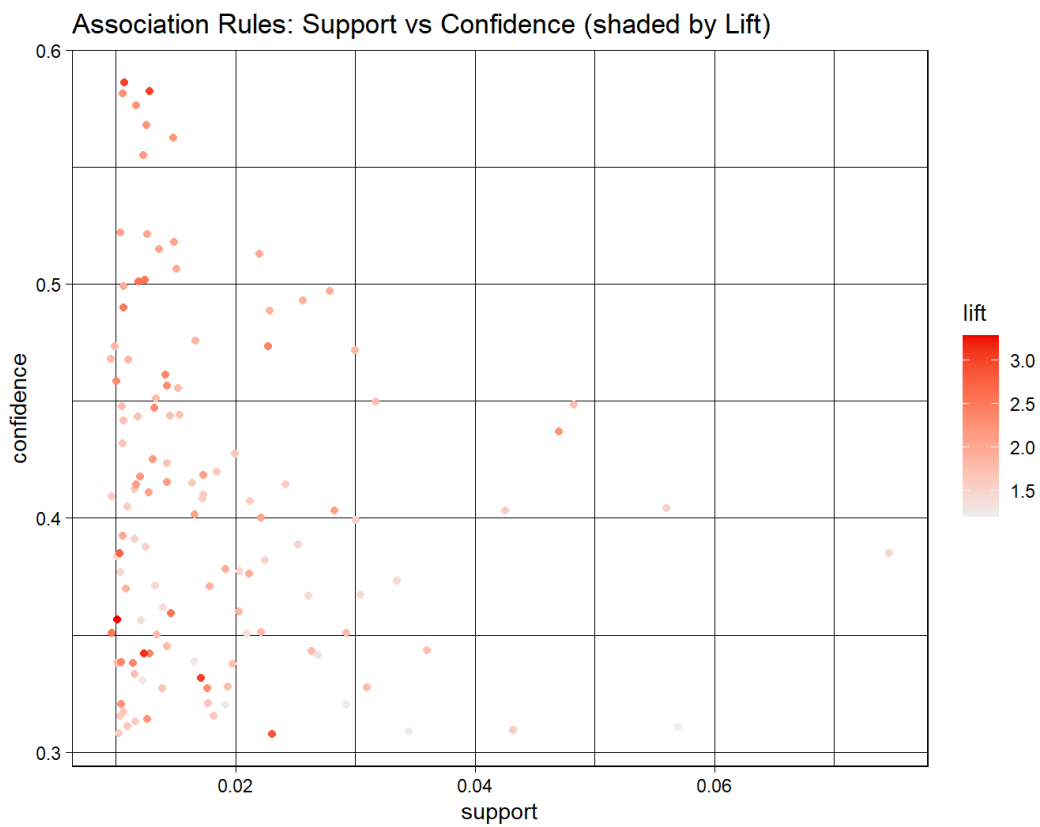


Figure 2: All 125 rules plotted by support and confidence, shaded by lift

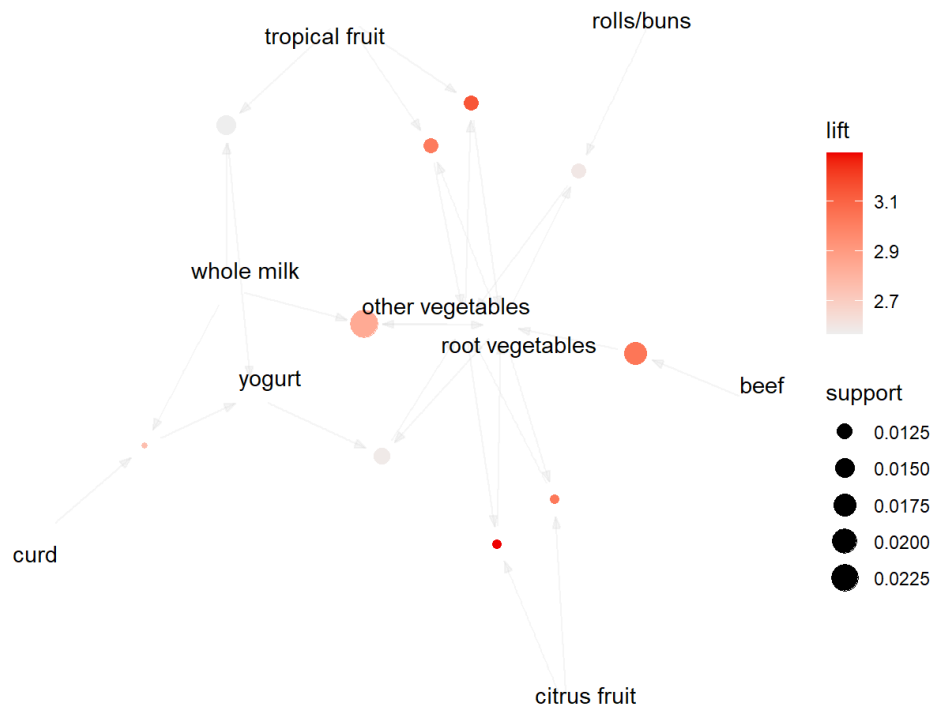


Figure 3: Network graph of the top 10 rules by lift node size reflects support, node/edge color reflects lift

### Answer

**a-b. Association rules generated:** At least 10 association rules, each reported with its antecedent, consequent, support, confidence, and lift, are listed in Table 3 (the top 10 by lift, drawn from all 125 rules meeting support  $\geq 0.01$  and confidence  $\geq 0.30$ ).

## 4 Understanding the Measures

Three rules from Table 3 are used to illustrate the three measures:

- **Rule 1:** {citrus fruit, root vegetables}  $\rightarrow$  {other vegetables} (support  $\approx 0.010$ , confidence  $\approx 0.586$ , lift  $\approx 3.03$ )
- **Rule 2:** {root vegetables, tropical fruit}  $\rightarrow$  {other vegetables} (support  $\approx 0.012$ , confidence  $\approx 0.585$ , lift  $\approx 3.02$ )
- **Rule 3:** {other vegetables}  $\rightarrow$  {whole milk} (support  $\approx 0.075$ , confidence  $\approx 0.387$ , lift  $\approx 1.51$ )

**Answer**

**a. Support:** Support is the proportion of *all* transactions that contain every item in the rule. For Rule 1, support  $\approx 0.010$  means citrus fruit, root vegetables, and other vegetables all appear together in about 1% of the 9,835 transactions.

**b. Confidence:** Confidence is, of the transactions that contain the antecedent, the proportion that also contain the consequent. For Rule 2, confidence  $\approx 0.585$  means that when a customer buys root vegetables and tropical fruit, they also buy other vegetables about 58.5% of the time.

**c. Lift:** Lift measures how much more likely the consequent is, given the antecedent, compared to its baseline purchase rate. For Rule 3, lift  $\approx 1.51$  means customers who buy other vegetables are about 51% more likely to also buy whole milk than a random customer a lift above 1 indicates a real positive association, not coincidence.

**5 Find Interesting Rules**

```
top_confidence <- sort(rules, by = "confidence", decreasing = TRUE)[1]
top_lift       <- sort(rules, by = "lift",     decreasing = TRUE)[1]
top_support    <- sort(rules, by = "support",  decreasing = TRUE)[1]

cat("Highest confidence rule:\n"); inspect(top_confidence)
cat("\nHighest lift rule:\n");   inspect(top_lift)
cat("\nHighest support rule:\n"); inspect(top_support)
```

Highest confidence rule:

| lhs                             | rhs                   | support    | confidence |
|---------------------------------|-----------------------|------------|------------|
| {citrus fruit, root vegetables} | => {other vegetables} | 0.01037112 | 0.5862069  |
| coverage                        | lift                  | count      |            |
| [1] 0.01769192                  | 3.029608              | 102        |            |

Highest lift rule:

| lhs                              | rhs                  | support    | confidence |
|----------------------------------|----------------------|------------|------------|
| {citrus fruit, other vegetables} | => {root vegetables} | 0.01037112 | 0.3591549  |
| coverage                         | lift                 | count      |            |
| [1] 0.02887646                   | 3.295045             | 102        |            |

Highest support rule:

| lhs                | rhs             | support    | confidence | coverage  | lift     |
|--------------------|-----------------|------------|------------|-----------|----------|
| {other vegetables} | => {whole milk} | 0.07483477 | 0.3867578  | 0.1934926 | 1.513634 |
| count              |                 |            |            |           |          |
| [1] 736            |                 |            |            |           |          |

**Answer**

**a. Highest confidence:** {citrus fruit, root vegetables}  $\rightarrow$  {other vegetables}, confidence  $\approx 0.586$  (58.6%).

**b. Highest lift:**  $\{\text{citrus fruit, other vegetables}\} \rightarrow \{\text{root vegetables}\}$ , lift  $\approx 3.30$ .

**c. Highest support:**  $\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$ , support  $\approx 0.075$  (7.5%).

**d. Are these the same rule?** No, these are three *different* rules. Support favors rules built on very common individual items (whole milk and other vegetables are simply purchased often, so any rule connecting them has high support), while confidence and lift favor rules involving less common but tightly associated items (like citrus fruit and root vegetables) that occur together far more than chance would predict, even though they don't occur in a large absolute number of transactions. A rule can be common but only weakly associated (high support, low lift), or rare but very strongly associated (low support, high lift/confidence) exactly what is observed here.

## 6 Interpret a Rule

**Selected rule:**  $\{\text{root vegetables, tropical fruit}\} \rightarrow \{\text{other vegetables}\}$  (support  $\approx 0.012$ , confidence  $\approx 0.585$ , lift  $\approx 3.02$ )

### Answer

**In simple language:** Customers who purchase root vegetables and tropical fruit together also tend to purchase other vegetables. This combination shows up in about 1.2% of all shopping trips, and when a customer buys root vegetables and tropical fruit, about 58.5% of the time they also pick up other vegetables.

**Strong or weak association?** This is a **strong association**. The confidence (58.5%) is well above the overall purchase rate of other vegetables ( $\approx 19\%$ ), and the lift of  $\approx 3.0$  means the combination is roughly three times more likely to include other vegetables than would be expected by chance. High confidence together with a lift well above 1 indicates a genuine, non-coincidental purchasing pattern rather than a weak or spurious link.

## 7 Business Application

Imagine acting as a supermarket manager using these results.

### Answer

**a. Two products to place close together:** Root vegetables and other vegetables. They co-occur frequently (support  $\approx 0.047$ ) and have a strong association (lift  $\approx 2.25$ ), so positioning them in adjacent produce sections would match natural shopping behavior and likely encourage additional add-on purchases.

**b. A possible product bundle:** A “fresh basics” bundle of root vegetables + tropical fruit + other vegetables, since customers who buy the first two very reliably ( $\approx 58.5\%$  of the time) also want the third a discounted bundle would reward behavior that already happens naturally and could nudge the remaining  $\approx 42\%$  to add it too.

**c. Example recommendation use:** An online grocery site or self-checkout kiosk could use a high-lift rule such as  $\{\text{other vegetables, whole milk}\} \rightarrow \{\text{root vegetables}\}$  to power a “customers

who bought this also bought...” recommendation panel, automatically suggesting root vegetables to any shopper with other vegetables and whole milk already in their cart, increasing basket size.

## 8 Understanding Association

### Answer

**a. Does an association rule prove causation?** No, an association rule does not prove causation. It only shows that items tend to be purchased in the same transaction more often than expected by chance. There could be a common underlying cause (e.g., both are ingredients in a popular recipe, or both are on promotion the same week) rather than one purchase causing the other. Establishing causation would require a controlled experiment (e.g., A/B testing shelf placement), not just observational transaction data.

**b. High confidence but low lift:** A rule can have high confidence but low lift when the consequent is already very popular on its own. For example, a rule ending in {whole milk} can easily have high confidence simply because most shoppers buy whole milk regardless of what else is in their cart the antecedent isn’t really driving the purchase, it’s just riding along with an item everyone buys anyway. Lift corrects for this by comparing confidence to the consequent’s baseline popularity.

**c. Lift greater than 1:** A lift value greater than 1 indicates a positive association: the items are purchased together more often than would be expected if purchased independently. The higher above 1, the stronger the association. A lift of exactly 1 means no association (statistical independence), and a lift below 1 means the items are negatively associated buying one makes the other less likely.

## 9 Conclusion

Association rule mining on the Groceries dataset confirmed that everyday staples whole milk, other vegetables, and rolls/buns dominate raw purchase frequency and drive the highest-*support* itemsets and rules simply by virtue of how often they are bought. However, ranking rules by *lift* surfaced a more actionable signal: fresh-produce combinations involving citrus fruit, root vegetables, tropical fruit, and beef show the strongest genuine associations, with lift values reaching 3.30. These findings translate directly into merchandising actions adjacent shelf placement, bundle promotions, and cart-based recommendations that a supermarket could deploy to increase basket size, while the analysis also demonstrates the important distinction between statistical association and causation that must be kept in mind when acting on rule-mining results.

## 10 Appendix: Full R Code

The complete R code used for this analysis, combining all steps from data loading through rule generation and visualization, is reproduced below.

```
# ---- Setup ----  
library(arules)  
library(arulesViz)  
library(knitr)
```

```
library(RColorBrewer)

# ---- Load Data ----
# Load from groceries.csv (basket-format, no header,
# variable number of items per row).
transactions <- read.transactions("groceries.csv", format = "basket",
                                  sep = ",", rm.duplicates = TRUE)

summary(transactions)

# ---- Question 1: Understand the Data ----
n_transactions <- length(transactions)
n_items <- ncol(transactions)
n_transactions
n_items

top10 <- sort(itemFrequency(transactions, type = "relative"), decreasing = TRUE)[1:10]
kable(data.frame(Item = names(top10), Support = round(as.numeric(top10), 4)))

itemFrequencyPlot(transactions, topN = 10, type = "relative",
                  main = "Top 10 Most Frequent Items",
                  col = brewer.pal(8, "Pastel2"))

# ---- Question 2: Frequent Itemsets ----
itemsets <- apriori(transactions, parameter = list(
  target = "frequent itemsets",
  support = 0.01,
  minlen = 2
), control = list(verbose = FALSE))

itemsets_sorted <- sort(itemsets, by = "support", decreasing = TRUE)
itemsets_df <- as(itemsets_sorted, "data.frame")
kable(head(itemsets_df, 10))

# ---- Question 3: Association Rules ----
rules <- apriori(transactions, parameter = list(
  support = 0.01,
  confidence = 0.3,
  minlen = 2
), control = list(verbose = FALSE))

length(rules)

rules_by_lift <- sort(rules, by = "lift", decreasing = TRUE)
rules_df <- as(rules_by_lift, "data.frame")
kable(head(rules_df, 10))

# Export full rule table for submission
write.csv(rules_df, "association_rules_output.csv", row.names = FALSE)

plot(rules, measure = c("support", "confidence"), shading = "lift",
     main = "Association Rules: Support vs Confidence (shaded by Lift)")
```

```
plot(head(rules_by_lift, 10), method = "graph",
     main = "Top 10 Rules by Lift - Network Graph")

# ---- Question 5: Find Interesting Rules ----
top_confidence <- sort(rules, by = "confidence", decreasing = TRUE)[1]
top_lift       <- sort(rules, by = "lift",      decreasing = TRUE)[1]
top_support    <- sort(rules, by = "support",  decreasing = TRUE)[1]

cat("Highest confidence rule:\n"); inspect(top_confidence)
cat("\nHighest lift rule:\n");   inspect(top_lift)
cat("\nHighest support rule:\n"); inspect(top_support)
```

*Methodology note: minimum support = 0.01 and minimum confidence = 0.30 were used for rule generation, with minlen = 2 to exclude trivial single-item rules. All tables and figures in this report were generated directly from the live R output of this script.*